Low-Cost Human Following Robot Using Shielded Dual-ESP32 Wi-Fi RSSI Sensing

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Methodology

The user carries a hotspot-enabled mobile device. Two ESP32 sensing nodes are mounted at the extreme ends of the horizontal frontal arm of the rover, with metallic sheets between them for EMI shielding, so the closely placed nodes cannot directly interfere with each other's signal sensing and cause errors. Digital filters (median, moving average) are built into the RSSI detection and movement algorithm to suppress cross-interference from the other node. The third ESP32, the main controller of the rover, receives the RSSI values from both sensing nodes over UART and decides the rover's movement. The difference between the values gives the direction of the user, and the controller steers toward it, applying a deadband with hysteresis and driving two BTS7960 drivers with PWM for smooth differential steering of four geared motors. A front ultrasonic/ToF sensor keeps a safe gap, and loss of signal triggers an automatic stop.

Proposed Block Diagram

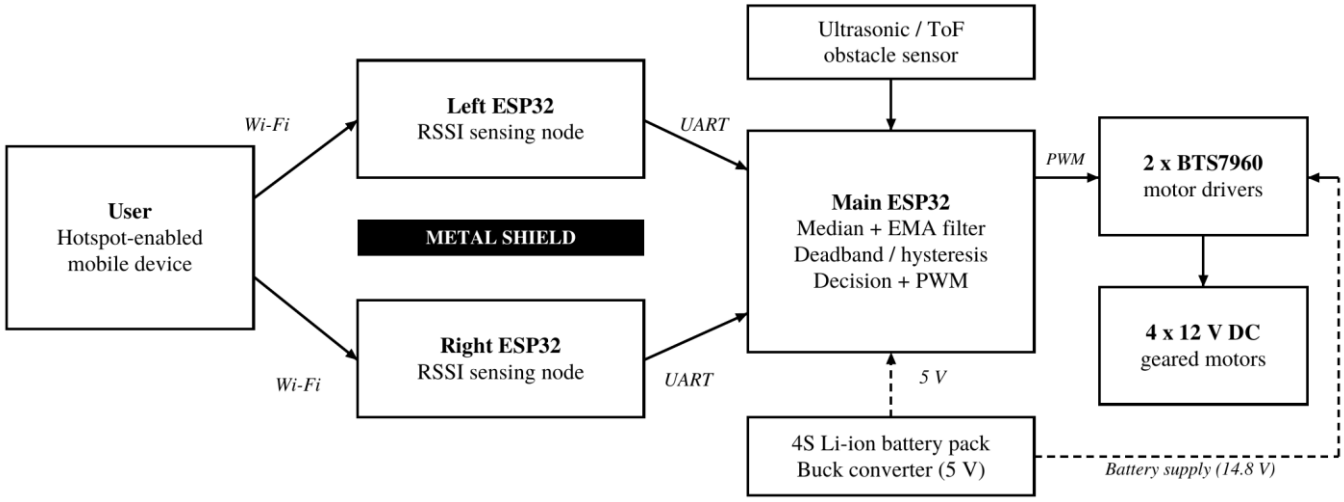


Fig. 1: Proposed system block diagram

Control Logic (Main ESP32)

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loop every 50 ms:
    L = filter(rssi_left);  R = filter(rssi_right)
    diff = L - R;  avg = (L + R) / 2
    if signal_lost or obstacle_near or avg > NEAR_TH: stop()
    else if |diff| > DEADBAND: steer toward stronger side (PWM ∝ |diff|)
    else if avg < FAR_TH: forward(PWM ∝ FAR_TH - avg)
    else stop()
```

Proposed Outcome and Real-World Implementation

Implementation: a hands-free load carrier for warehouses, farms, hospitals, campuses and airports. It also helps elderly persons and persons with limited mobility to carry bags while walking. Since the human chooses the route, no mapping or vision is needed, keeping cost near Rs. 5,000.

Expected outcome: a working prototype that follows a walking user at roughly 0.5-2.5 m indoors with stable turning and safe stopping.

Future scope: wearable ESP-NOW tag, CSI sensing, on-device TinyML for direction classification, and multi-robot fleets with IoT dashboards.